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Write-up (Homework 1)

For this assignment, I was supposed to complete EDA on an obfuscated dataset. Upon loading the dataset into pandas, I noted that there were 2000 observations and 552 features. The dataset includes a mix of floats, ints and object columns. This indicates that there is a mix of nominal, ordinal, and numerical data. An example of a nominal feature in the dataset is MS_enc. An example of an ordinal feature in the data is SMIF. Based on my interpretation of what the data could possibly be, it is a series of compounds and their properties. Grouping the data by MS_Enc to identify the most common compound, it appears to be PPENPINEAPPLE42 at a little over 1400 observations.

To preprocess the data, I checked for na values by using the .isna() function, checking the minimum value of each column and then the maximum. I didn't see any stand out issues so I then moved on to one hot encoding the column MS_enc column. It was after encoding and then developing the correlation matrix I received an error stating, "Computation failed". That then told me, those were the NaN values in the dataset. Going back, I changed all of those values to NaN and then dropped those observations from the dataset. That left me with 1805 observations.

It was during this process I realised I need to encode the SMIF column as well. Once that was done, I generated a heatmap, to better visualize the data. There are many outliers in this data set. Due to the massive amount of features it was very unclear, so I created smaller dataframes that aim to highlight the variables with the strongest and weakest correlation, making sure to ignore duplicates and self correlations. There are many variables that have an extremely weak correlation (less than 0.1, in terms of absolute values) but there are none that

have a correlation value of 0. There are 9 pairs of variables that possess a correlation value of 1. The feature with the highest correlation to experimental_property is SLogP at 0.427319.

After that, I proceeded with PCA. For PCA to work, I had to filter out columns so I am only working with those that are strictly numerical. I did that by filtering out columns that are floats. I used the experimental_property column as the target (or y) and the other columns as X. I initially tried using 3 components to fit the data and used matplotlib to visualize. However, in one of the observations, the target value is NaN. So I switched to 2, but that did not work as there is still 1 point that has a target value of NaN.

Using the explain_variance_ratio, I got the values 0.33344061 and 0.08347055. What this means is that for the first component, pca can explain approximately 33% of the variance. This is very low, so it shows that using PCA to explain the variance in the data is not possible and another approach must be used. Nonetheless, I then used the viridis color map for my PCA visualization to show which data point can be explained by which component. The plot that is produced shows that many of these points are clumped together with no semblance of a line, hence we must use non-linear dimensionality to analyze the data.

I then attempted to use t-SNE to reduce the dimensionality. Initially, I thought I would need to use cuml for it to work, however, I learnt that sci-kit learn has a package that allows me to perform t-NSE dimensionality reduction. Using that package I created another visualization. However, there is no clear differentiation between targets.